

Branch Specialization Checkpoint

All-Layout Local-Weight Sweep

Branch Specialization Project

2026-05-22

Checkpoint reason: full 100-model grid refines the capacity-attractor mechanism.

Project question: Does structural heterogeneity lead to stable functional specialization?

Previous result: for one layout, reducing local-task pressure let induction use the 64-dim head.

This checkpoint asks whether that result generalizes across all four 64-dim placements:

16, 32

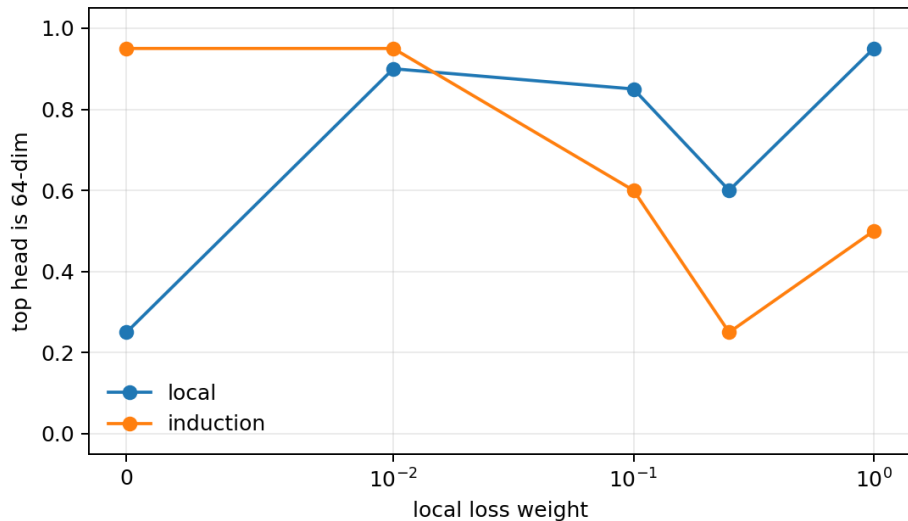
16, 32

64, 16

32, 64

- Task has two roles: local copy and induction-style continuation.
- Local weight swept: 0.00, 0.01, 0.10, 0.25, 1.00.
- Induction weight fixed at 1.00.
- Four heterogeneous layouts, five seeds each: 100 models total.
- Role score: single-head causal ablation loss delta.
- Main readout: whether the role's top causal head is 64-dim.

Aggregate Evidence

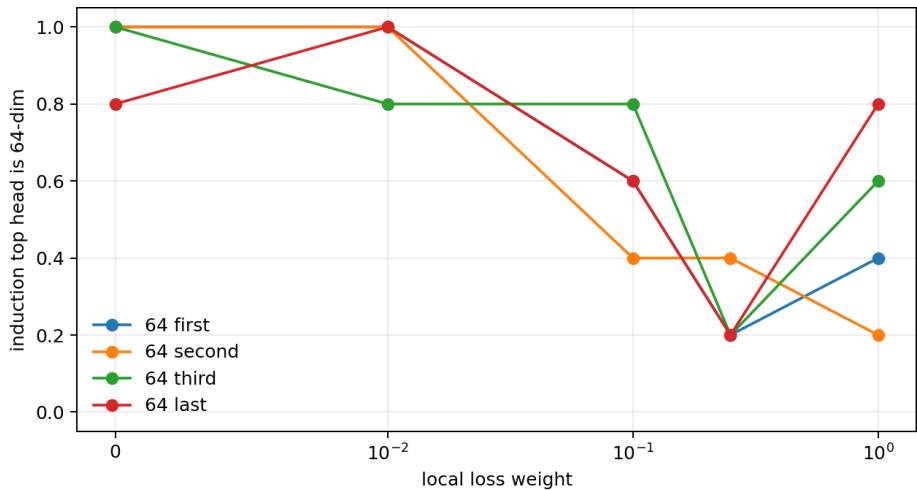


Aggregate Table

Local weight	Models	Local acc.	Induction acc.	Local top 64	Induction top 64
0.00	20	0.2006	0.9979	0.25	0.95
0.01	20	0.9955	0.9979	0.90	0.95
0.10	20	0.9998	0.9983	0.85	0.60
0.25	20	1.0000	0.9986	0.60	0.25
1.00	20	1.0000	0.9993	0.95	0.50

Induction top-64 rate changes from 19/20 at low local pressure to 5/20 at local weight 0.25.

Layout-Specific Evidence



The pressure effect generalizes, but the curve is not strictly monotonic and remains layout-sensitive.

Supported:

- The 64-dim head is a real attractor for induction when local pressure is absent or tiny.
- Local pressure changes role allocation among structural slots.
- Heterogeneity stabilizes slot formation across seeds.

Not supported:

- A fixed semantic taxonomy where large heads always mean induction/global.
- A one-variable monotonic law from local weight to induction displacement.

Best current mechanism:

Structural heterogeneity creates capacity-attractor slots; role allocation among those slots is pressure-sensitive, layout-sensitive, and optimization-basin-sensitive.

This supports functional specialization stability in the toy setting. It does not yet establish functional modularity.

Continue Phase 3, but with the narrowed claim.

Next decisive experiment:

- Add two-attractor layouts, e.g. $[16, 48, 48, 16]$, $[48, 16, 48, 16]$, $[48, 48, 16, 16]$.
- Test whether local and induction split more cleanly when there are two medium-large heads.
- If displacement weakens, it directly supports the capacity-attractor competition mechanism.

Sources and Provenance

- Report: `doc/phase3_toy_competition_all_layout_weight_sweep.md`
- Training script: `scripts/toy_competition_head_dim_intervention.py`
- Analysis script: `scripts/analyze_competition_all_layout_weight_sweep.py`
- Combined analysis: `results/phase3_toy_competition_all_layout_weight_sweep/`
- Deck data snapshot: `presentations/2026-05-22-1135-all-layout-weight-sweep/data/`